S01 SOUND MAPPING | P

Intent : Incorporate strategic planning required to prevent issues of acoustic disturbance from various sources of noise.

Summary : This WELL feature requires projects to strategize an acoustical plan that identifies sources of noise that can negatively impact interior spaces.

Issue : Architectural trends, such as open workspaces, lightweight construction and exposed finishes or HVAC foster acoustically uncomfortable environments. [1–4] When noise from internal or external sources is increased in a space, occupants have been found to be easily distracted, less productive and susceptible to burnout.[5–9] In office environments, employees care about privacy and collaboration.[8,10] In one study from the United Kingdom, 99% of employees reported that their concentration was impaired by poor acoustical comfort in the workplace.[7] Similar conditions are reported in workplaces worldwide. [5,8,11–13] Some reports have shown that occupants are less likely to help others under high noise conditions, reducing collaboration in the workplace.[6]

Solutions : To address noise, floor plans should be designed with intent and use in mind.[14] For example, a typical office can be categorized into four key types of programming: spaces for concentration, collaboration, socialization and learning.[15] Location is important, since noise from social or collaborative spaces impacts spaces intended for concentration or learning.[15,16] This approach can be implemented in any space type that incorporates spaces of socialization and recreation alongside areas for task-centric work or learning.[15] These spaces can then be described as loud, quiet and mixed spaces to better assess the impact of sound on sensitive, quiet locations for concentration, learning or recreation.

PART 1 LABEL ACOUSTIC ZONES

For All Spaces:

The project provides the following:

a. A floor plan or other design document showing the following acoustic zones throughout the project:
   1. Loud zone: includes areas intended for loud equipment or activities (e.g., mechanical rooms, AV/IT closets, kitchens, fitness rooms, social spaces, recreational rooms, music rooms).
   2. Quiet zone: includes areas intended for concentration, wellness, rest, study and/or privacy (e.g., restorative spaces, lactation rooms, nap rooms).
   3. Mixed zone: includes areas intended for learning, collaboration and/or presentation (e.g., auditoriums, classrooms, breakout spaces).
   4. Circulation zone: includes occupiable areas not intended for regular occupancy (e.g., hallways, egress, atria, stairs, lobbies)
   5. Not applicable zones: includes other areas without significant sources of sound (e.g., storage rooms, janitor rooms, coat closets) that are not regularly occupied.

b. A plan for reprogramming or mitigating sound transmission between loud zones that border quiet zones (if any).

PART 2 PROVIDE ACOUSTIC DESIGN PLAN

For All Spaces:

The project provides one of the following:

a. A plan developed by the project team and/or project owner that outlines acoustical solutions and a timeline for implementation with a focus on managing acoustical comfort, background noise, speech privacy, reverberation time and/or impact noise within the project boundary.

b. A detailed report from a professional in acoustics that describes existing conditions, recommended solutions and measurement results with a focus on managing background noise, speech privacy, reverberation time and/or impact noise within the project boundary. These measurements are not required to adhere to the Performance Verification Guidebook requirements for on-site testing.

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